

Topic 6: Google Gemini and Google Workspace AI

6.1 Google Gemini Overview

- Gemini is Google's family of multimodal AI models: Ultra, Pro, Flash, and Nano (for on-device use)
- Gemini.google.com: consumer chatbot, free with a Google account; Google One AI Premium (\$20/month) for Gemini Advanced
- Gemini is natively multimodal -- designed from the ground up to process text, images, audio, and video together
- Deep Google integration: performs real-time Google Search, accesses Google Drive, Docs, and Gmail with permission
- Try it: <https://gemini.google.com> | Developer playground: <https://aistudio.google.com>

6.2 Gemini in Google Workspace

- Gmail: 'Help me write' drafts an email from bullet points; 'Summarise this thread' condenses long chains
- Google Docs: draft full documents from prompts, rewrite or extend selected sections, summarise documents
- Google Slides: 'Create a presentation about renewable energy trends for a non-technical audience' generates slides with images
- Google Sheets: natural language formulas and data analysis ('What are the top 3 trends in this dataset?')
- Google Meet: real-time captions, translated captions in 70+ languages, automatic post-meeting summaries

6.3 NotebookLM -- AI Research Assistant

- NotebookLM is a free Google tool that lets you upload your own documents and ask questions about them
- Supported sources: PDFs, Google Docs, YouTube video URLs, web URLs, and audio files
- Responses are grounded in your uploaded sources -- citations link back to exact passages
- Audio Overview: generates a realistic podcast-style conversation between two AI hosts discussing your documents
- Use cases: studying from textbooks, synthesising research papers, preparing for meetings, understanding contracts
- Try it: <https://notebooklm.google.com> (free, no subscription required)

6.4 Google AI Studio and the Gemini API

- Google AI Studio (aistudio.google.com): free browser-based tool to experiment with Gemini models
 - Test prompts, adjust parameters (temperature, top-p), and automatically get Python or JavaScript API code
 - Generous free tier: Gemini 1.5 Flash offers 15 requests per minute at no cost -- good for learning and prototyping
 - Multimodal testing: upload images, audio clips, or video alongside text prompts to test combined inputs
 - One-click code export: any prompt you build in AI Studio generates ready-to-run API code
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